Module 09

Data Preparation & Quality for Generative AI Applications

Why data quality decides AI reliability, and how to build trustworthy RAG systems through rigorous data preparation

GENAI, NLP AND PROMPT ENGINEERING

MODULE 09

Building on What We've Learned

01

Modules 04–05: Retrieval + RAG Pipeline

We explored how retrieval-augmented generation works and built the foundational pipeline for connecting language models to external knowledge bases.

02

Module 06: Evaluation and Metrics

We learned to measure system performance using precision, recall, and faithfulness metrics to ensure our AI produces reliable outputs.

03

Module 08: NLP Prompt Patterns

We mastered techniques for extraction, classification, and redaction—essential patterns for controlling how models process and present information.

04

Today: Data Quality Foundations

Now we focus on the critical foundation: clean data + safe data = accurate retrieval + faithful answers. Without quality data, even the best models will fail.

Why Data Quality Matters in Generative AI

Data quality is not just a technical concern—it's the foundation of trustworthy AI systems. Poor data quality cascades through every stage of your RAG pipeline, undermining reliability and user trust.

Hallucinations and Wrong Answers

Poor-quality data directly causes models to generate false information. When source documents contain errors or ambiguities, the model cannot distinguish truth from noise.

Inflated Performance Metrics

Duplicates create artificial confidence by making retrieval "too easy." Your evaluation scores look great, but they don't reflect real-world performance.

Amplified Bias

Biased sources lead to biased outputs. If your training data overrepresents certain perspectives or demographics, your AI will replicate and amplify those biases.

Reduced Trust and Accountability

Missing documentation makes it impossible to trace where information came from or validate its accuracy. This destroys accountability in production systems.

Small Dataset Vulnerability

With limited data, **every single record matters**. A single bad document can significantly skew results, making rigorous cleaning absolutely essential.

The RAG Data Lifecycle

Understanding how data flows through your system helps identify where quality issues emerge and where interventions have the greatest impact.

1. **Collection**
   Gather source documents from approved repositories
2. **Preprocessing**
   Clean, normalize, and structure raw data
3. **Indexing**
   Create searchable embeddings and metadata
4. **Retrieval**
   Find relevant documents for user queries
5. **Generation**
   Produce answers grounded in retrieved context
6. **Evaluation**
   Measure accuracy and refine the pipeline

Critical insight: Each stage must preserve accuracy, safety, and traceability. If one stage fails, output quality drops across the entire system. Quality control is not optional—it's foundational.

Lifecycle Objectives: Quality Goals

These five objectives guide every data decision you make. They ensure your RAG system produces reliable, trustworthy, and safe outputs for all users.

Accuracy

All content must be correct and verified against authoritative sources. Inaccurate data leads directly to hallucinations and user distrust.

Coverage

Include all key topics and domains, not just popular sections. Incomplete coverage creates blind spots where your system cannot help users.

Fairness

Avoid skewed sources or language dominance. Ensure diverse perspectives are represented and no demographic is systematically excluded.

Transparency

Maintain clear source records and processing logs. Users and auditors must be able to trace where information originated and how it was transformed.

Safety

Exclude all private, sensitive, or harmful data. Safety violations can cause legal liability and severe harm to individuals.

Data Sources: What's Allowed vs. Forbidden

✓ Allowed Sources

Use only approved public documents that are intended for wide distribution and contain no sensitive information:

- Official institutional handbooks and policy documents
- Published notices and announcements
- Public FAQs and knowledge base articles
- Academic course catalogs and syllabi
- Publicly available research papers and reports

☐ Forbidden Sources

Never include personal or sensitive records that could violate privacy or expose confidential information:

- Student records or grades
- Personal correspondence or emails
- Financial or medical information
- Internal-only communications
- Documents marked confidential

> **Golden Rule:** If a document cannot be publicly displayed on a website or bulletin board, do not include it in your dataset. When in doubt, consult with data governance teams.

Sampling & Dataset Construction

Sampling is the strategic selection of a representative subset from a larger corpus. Rather than using everything available, thoughtful sampling creates more balanced, efficient, and maintainable datasets.

1

Balanced Topic Representation

Ensure all important topics appear proportionally. Avoid datasets dominated by a single category like "Admissions only" which creates blind spots in other areas.

2

Reduced Redundancy

Eliminate repetitive content that inflates dataset size without adding new information. Less redundancy means faster retrieval and more diverse results.

3

Faster Retrieval Performance

Smaller, well-curated datasets reduce computational overhead and improve response times. Quality beats quantity in retrieval systems.

4

Prevents Overfitting

Diverse sampling helps your model generalize better across different query types rather than memorizing patterns from overrepresented topics.

Stratified Sampling in Practice

Stratified sampling divides your corpus into meaningful categories and ensures each category is properly represented. This technique prevents category dominance and ensures comprehensive coverage.

Step 1: Define Categories

Identify natural divisions in your domain: Admissions, Hostel, Library, Exams, Student Services

Step 2: Set Targets

Allocate records proportionally or equally depending on use case requirements

Step 3: Diversify Within

Include document variety: notices, policies, FAQs, announcements

Step 4: Validate Balance

Verify coverage and adjust if any category is underrepresented

Example: University Chatbot Dataset

Category	Target Records	Document Types	Priority Topics
Admissions	40 records	Policy docs, FAQs, notices	Requirements, deadlines, process
Hostel	30 records	Handbooks, announcements	Allocation, rules, facilities
Library	30 records	Guides, policies	Hours, borrowing, resources
Exams	40 records	Schedules, regulations	Dates, rules, procedures
Student Services	30 records	Service descriptions, FAQs	Counseling, health, career

This stratified approach ensures balanced coverage across all critical student-facing services while maintaining document diversity within each category.

Deduplication: Why It's Critical

Duplicate records are one of the most insidious data quality issues in RAG systems. They create false confidence, skew evaluation metrics, and waste computational resources.

1

False Confidence in Performance

When the same content appears multiple times, your system appears to perform better than it actually does. Duplicates make retrieval artificially easy, masking real weaknesses.

2

Dominated Retrieval Results

Repeated content monopolizes top results, preventing diverse and comprehensive answers. Users see the same information repeatedly instead of varied perspectives.

3

Inflated Evaluation Scores

Evaluation becomes meaningless when test queries repeatedly retrieve identical duplicates. Your metrics look great but don't reflect real-world capability.

Impact Comparison

Scenario	With Duplicates	After Deduplication
Top-5 retrieval diversity	2 unique documents	5 unique documents
Evaluation accuracy	95% (inflated)	78% (realistic)
User satisfaction	Low (repetitive answers)	High (diverse insights)
Index size	500 MB	320 MB (36% reduction)

Deduplication Methods: Manual + Automated

Choose your deduplication approach based on dataset size and available resources. Manual methods work for small datasets, while automated techniques scale to production environments.

Manual Approaches (Small Datasets)

For datasets under 500 records, manual review combined with spreadsheet tools provides high accuracy:

- **Excel Remove Duplicates:** Built-in feature for exact match detection
- **Conditional Formatting:** Highlight similar text for visual inspection
- **Sort & Compare:** Sort by title or content, manually identify near-duplicates
- **Hash Comparison:** Generate content hashes, flag exact matches

Automated Techniques (Larger Datasets)

For datasets exceeding 500 records, automated similarity detection becomes essential:

- **Cosine Similarity:** Compare TF-IDF vectors to find similar documents
- **Embedding Distance:** Use sentence transformers to detect semantic duplicates
- **Fuzzy Matching:** Levenshtein distance for near-text matches
- **MinHash LSH:** Efficient approximate matching for massive datasets

Rule of Thumb: Set similarity threshold ≥ 0.90 to treat documents as duplicates. Lower thresholds (0.80-0.89) flag for manual review. This balances precision with recall in duplicate detection.

```
# Simple Python deduplication example
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.metrics.pairwise import cosine_similarity

# Calculate similarity matrix
vectorizer = TfidfVectorizer()
tfidf_matrix = vectorizer.fit_transform(documents)
similarity = cosine_similarity(tfidf_matrix)

# Flag duplicates above threshold
duplicates = similarity >= 0.90
```

Leakage Prevention: Train/Test Separation

Data leakage occurs when the same or highly similar content appears in both your retrieval corpus and evaluation set. This creates unrealistically high accuracy scores that don't reflect true system capability.

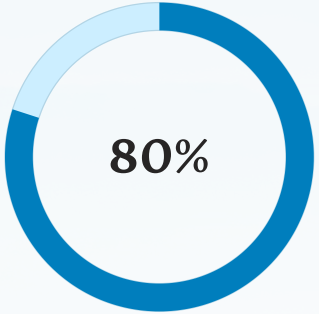
1. **Problem: Leakage**
   Evaluation queries retrieve their own answers directly from the corpus, making accuracy artificially perfect.

2. **Consequence**
   You believe your system works great, but it fails on real user queries because it never learned to generalize.

3. **Solution: Split First**
   Separate data into corpus and evaluation sets **before** any processing or indexing begins.

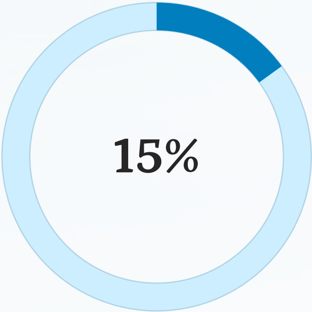
4. **Build Second**
   Only after splitting, proceed to clean, preprocess, and index each set independently.

Recommended Split Ratios



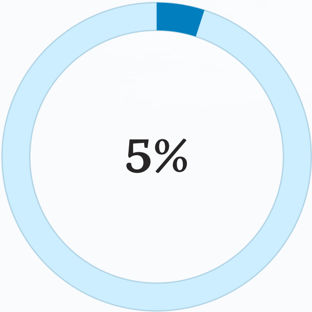
Retrieval Corpus

Documents indexed for production queries



Validation Set

For tuning hyperparameters and prompts



Test Set

Final evaluation, never touched during development

> Critical rule: Once you split, the test set becomes sacred. Never inspect, adjust, or leak information from it until final evaluation. Contamination destroys validity.

Dataset Documentation: Data Cards

Data cards are structured documentation that explains what your dataset contains, how it was created, and what limitations it has. They're essential for transparency, accountability, and ethical AI development.

Why Data Cards Matter

- **Enables Review:** Instructors and stakeholders can verify ethical compliance
- **Ensures Reproducibility:** Future developers understand data provenance
- **Documents Bias:** Known limitations and skews are explicitly stated
- **Supports Auditing:** Provides trail for regulatory compliance
- **Builds Trust:** Transparency increases confidence in AI systems

Mandatory Fields

Field	Description
Purpose	Intended use case and goals
Source	Origin of all documents
Categories	Topic breakdown and distribution
Privacy Review	Confirmation of PII removal
Bias Notes	Known limitations and skews
Collection Date	When data was gathered
Preprocessing	Cleaning steps applied
License	Usage rights and restrictions

Example Data Card

- **Name & Purpose:** University Student Services Chatbot Corpus v1.0, for training a RAG chatbot on student inquiries (admissions, housing, library, exams).
- **Source & Date:** Compiled from public university websites, handbooks, and FAQs; collected January 2024.
- **Content Breakdown:** 170 documents across Admissions, Hostel, Library, Exams, and Student Services.
- **Considerations:** All PII redacted. Biased towards traditional undergraduate policies; graduate/international student info underrepresented.

Cleaning & Preprocessing Tasks

Raw data is messy. Systematic preprocessing transforms chaotic source documents into clean, consistent, and trustworthy training data.

1

Remove HTML and Noise

Strip HTML tags, JavaScript, CSS, and formatting artifacts. Extract only semantic content.

2

Fix Encoding and Punctuation

Correct mojibake (encoding errors), normalize quotes/dashes, fix broken characters.

3

Standardize Capitalization

Apply consistent title case for headings, sentence case for body text. Fix ALL CAPS sections.

4

Normalize Terminology

Ensure consistent use of technical terms. Map synonyms to canonical forms (e.g., "dorm" → "residence hall").

5

Redact Sensitive Data

Replace all PII with [REDACTED] tokens. Never delete—redaction maintains document structure for transparency.

6

Standardize Dates

Convert all dates to single format (e.g., YYYY-MM-DD). Resolve ambiguities like "3/4/23".

Before and After Example

Before Cleaning

<div>ADMISSION DEADLINE is
03/15/23!!!Contact john.doe@uni.edu
for   questions.</div>

After Cleaning

Admission deadline is 2023-03-15.
Contact [REDACTED] for questions.

Impact: Clean data directly improves retrieval accuracy by reducing noise and ensuring semantic consistency across the corpus.

Quality Evaluation Checkpoints

Systematic validation ensures your dataset meets quality standards before deployment. Use these checkpoints at every stage of data preparation.

Coverage Check

Verify all critical topics are represented. Calculate category distribution and flag gaps.

Redundancy Check

Run deduplication analysis. Ensure similarity threshold caught all near-duplicates.

Clarity Check

Sample documents for readability. Verify consistent formatting and terminology.

Privacy Check

Scan for PII using regex patterns and manual review. Confirm all identifiers redacted.

Audit Trail

Document all checks in audit checklist. Maintain transparency for reviewers.

Validation Metrics

Metric	How to Measure	Target
Category Balance	Std dev of category sizes	< 20% variation
Duplicate Rate	% similar pairs > 0.90	< 2%
PII Leakage	Regex scan hits	0 instances
Document Completeness	% docs with all required fields	100%
Readability Score	Flesch Reading Ease	> 60

Quality Audit Checklist

- **Data Sourcing & Preparation Integrity:** Verified all source documents, completed deduplication and date normalization, and ensured train/test split has no leakage.
- **Quality & Privacy Controls:** Confirmed PII redaction via regex and manual spot-check, and ensured category balance meets distribution targets.
- **Final Validation & Documentation:** Completed data card with mandatory fields and validated the final dataset by a second reviewer.

Debrief & Homework

Debrief & Share

Reflect on today's learning with your team. Discuss these key questions:

1. **Top 3 Data Issues:** What are the most critical data quality problems you encountered or learned about today?
2. **Deduplication Process:** How did you identify and remove duplicates? What threshold did you use and why?
3. **Leakage Risk Example:** Describe one scenario where train/test leakage could occur and explain your prevention strategy.
4. **Data Cards & Trust:** Why are data cards essential for transparency, grading, and building trustworthy AI systems?

Homework Briefing

Task (Pairs): Prepare a 25-record mini dataset for a chatbot

- **Clean text:** Remove HTML artifacts, fix encoding, normalize dates
- **Redact identifiers:** Replace all PII with [REDACTED] tokens
- **Deduplicate:** Use similarity ≥ 0.90 threshold to identify duplicates
- **Stratify:** Organize into 5 categories with 5 records each
- **Create Data Card:** Document purpose, source, privacy review, bias notes

Submission Requirements:

- Clean CSV file (25 records, 5 categories)
- Deduplication audit sheet showing similarity scores and decisions
- Completed Data Card with all mandatory fields

Due: Before next class session. Submit via course portal.

Thank You!

You've learned the foundational skills for preparing high-quality data that powers reliable RAG systems. Clean data is the bedrock of trustworthy AI.

What We Covered

Data quality principles, lifecycle management, sampling strategies, deduplication, leakage prevention, and comprehensive documentation

Key Takeaway

Quality data = Accurate retrieval + Faithful answers

Coming Next: Module 10

Shipping a Demo: From Notebook to Interactive Application

We'll take your RAG system from experimental code to a production-ready web application. Learn deployment strategies, interface design, user experience patterns, and how to ship AI products that users love.

DEPLOYMENT | WEB APPS | USER EXPERIENCE